2 Chiziqli tenglamalar sistemasi

1-to'plam. *O'rniga qo'yish usuli bilan (x, y) ni toping.*

1. $y = x + 3, \quad 2x + y = 18$

2. $y = 2x, \quad x + y = 15$

3. $y = 3x - 4, \quad x + y = 12$

4. $x = y + 5, \quad 2y + x = 17$

5. $y = -x + 10, \quad 3x - y = 2$

6. $y = 4x - 3, \quad 2x + y = 21$

2-to'plam. *Qo'shish (eliminatsiya) usuli – ba'zilarida avval ko'paytirish kerak.*

1. $x + y = 9, \quad x - y = 3$

2. $2x + y = 13, \quad x - y = 2$

3. $3x + 2y = 17, \quad x - 2y = -5$

4. $2x + 3y = 13, \quad 4x - y = 5$

5. $5x + 2y = 19, \quad 3x - 2y = 5$

6. $x + 3y = 14, \quad 2x + y = 13$

3-to'plam. *Aralash – qaysi usul qulayroq, shuni tanlang.*

1. $y = 3x - 5, \quad 2x + 3y = 18$

2. $4x - y = 10, \quad 2x + y = 14$

3. $x + y = 20, \quad x - y = -6$

4. $3x + y = 7, \quad y = x - 5$

5. $2x + 5y = 22, \quad x - y = 4$

6. $6x - 2y = 24, \quad 3x + y = 18$

4-to'plam. *Har bir sistemani tasniflang:* ***bitta yechim*** *(uni toping),* ***yechimsiz****, yoki* ***cheksiz ko'p yechim****.*

1. $3x + 6y = 12, \quad x + 2y = 4$

2. $x + 2y = 5, \quad 2x + 4y = 9$

3. $x - 3y = 1, \quad 2x - 6y = 2$

4. $5x + y = 10, \quad x - y = 2$

k ning qiymatini toping:

5. $2x + ky = 6$ va $4x + 8y = 20$ sistemasi **yechimga ega bo'lmasligi** uchun k =?

6. $x + 3y = 6$ va $2x + ky = 12$ sistemasi **cheksiz ko'p yechimga ega bo'lishi** uchun k =?

5-to'plam. *Sistemani yeching, so'ng aynan so'ralgan kattalikni toping – shunchaki x yoki y emas.*

1 $x + y = 14,\ x - y = 4 - xy$ ni toping.

2 $2x + y = 11,\ x - y = 1 - x + y$ ni toping.

3 $x + 2y = 16,\ x - y = 1 - 2x - y$ ni toping.

4 $3x + y = 19,\ x + y = 9 - x - y$ ni toping.

5 $x + y = 13,\ 2x - y = 8 - x^2 - y^2$ ni toping.

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1 $x + y = 8,\ x - y = 2$ ifodasida ko'rsatilgan yechimda xato bor:

"Qo'shib: $2x = 8 - 2 = 6$, *demak* $x = 3,\ y = 5$."

Xato va to'g'ri javob: .....

2 $2x+y = 7,\ y = 3x-8$ ifodasida ko'rsatilgan yechimda xato bor:

"$2x + 3x - 8 = 7$, *demak* $5x = 7,\ x = \frac{7}{5}$."

Xato va to'g'ri javob: .....

3 $4x + 2y = 8,\ 2x + y = 5$ ifodasida ko'rsatilgan yechimda xato bor:

"Ikkala tenglama ham bitta chiziqni ifodalaydi, shuning uchun cheksiz ko'p yechim bor."

Xato va to'g'ri javob: .....

4 $3x-y = 5,\ x+y = 11$ ifodasida ko'rsatilgan yechimda xato bor:

"Qo'shib: $4x = 16,\ x = 4$; *so'ng* $y = 11 + 4 = 15$."

Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1 $2x + 3y = 7, \quad 5x - 2y = 8$

2 Ikki sonning yig'indisi 24, ayirmasi 8 ga teng. Bu ikki son nechaga teng?

3 $3x + ky = 10$ va $x + 2y = 3$ sistemasi **yechimga ega bo'lmasligi** uchun k =?

4 $4x + 3y = 25, \quad 2x - y = 5$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$(5, 8)$	2	$(5, 10)$
	3	$(4, 8)$	4	$(9, 4)$
	5	$(3, 7)$	6	$(4, 13)$
2-to'plam	1	$(6, 3)$	2	$(5, 3)$
	3	$(3, 4)$	4	$(2, 3)$
	5	$(3, 2)$	6	$(5, 3)$
3-to'plam	1	$(3, 4)$	2	$(4, 6)$
	3	$(7, 13)$	4	$(3, -2)$
	5	$(6, 2)$	6	$(5, 3)$
4-to'plam	1. Cheksiz ko'p yechim (2-tenglama 1-tenglamaning aynan $\frac{1}{3}$ qismi).			
	2. Yechimsiz (koeffitsient nisbati ikkala marta ham 2, lekin $5\times2 = 10 \neq 9$).			
	3. Cheksiz ko'p yechim (2-tenglama 1-tenglamaning aynan 2 barobari).			
	4. Bitta yechim: $(x, y) = (2, 0)$.			
	5. $k = 4$			
	6. $k = 6$			
5-to'plam	1. $xy = 45$ $(x = 9, y = 5)$			
	2. $x + y = 7$ $(x = 4, y = 3)$			
	3. $2x - y = 7$ $(x = 6, y = 5)$			
	4. $x - y = 1$ $(x = 5, y = 4)$			
	5. $x^2 - y^2 = 13$ $(x = 7, y = 6)$			
6-to'plam	1. Xato – $8 + 2$ o'rniga $8 - 2$ ishlatilgan. To'g'ri: $2x = 10$, $x = 5$, $y = 3$.			
	2. Xato – -8 had o'ng tomonga ko'chirilmagan. To'g'ri: $5x-8 = 7$, $x = 3$, $y = 1$.			
	3. Xato – konstantalar tekshirilmagan: nisbat mos kelmaydi (8 vs 10) – demak **yechimsiz**, cheksiz ko'p emas.			
	4. Xato – ishora: $y = 11 - 4 = 7$ bo'lishi kerak edi, $(4, 7)$.			
7-to'plam	1	$(2, 1)$	2. Sonlar: 16 va 8	
	3. $k = 6$			
	4	$(4, 3)$	–	–

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram